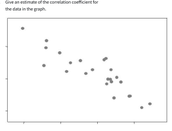
Due Mar 30, 11:59 PM CEST

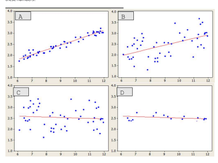
1. Give an estimate of the correlation coefficient for the data in the graph. 1 point



- -0.2
- -0.75
- 0.8
- -1

2. You perform a regression and find a p-value of 0.550 and an R-squared of 0.7%.

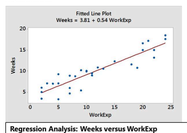
Which one of the graphs belongs to these numbers? 1 point



- A.
- B.
- C.
- D.

3. A regression is performed to determine the effect of work experience (WorkExp, in years) on the number of weeks it takes unemployed people to find a new job (Weeks).

Based on the outputs presented, what is the correct conclusion? 1 point



Regression Analysis: Weeks versus WorkExp

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The regression equation is
Weeks = 3.81 + 0.54 WorkExp

S = 1.76551   R-Sq = 83.1%   R-Sq(adj) = 82.5%

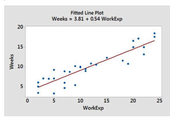
Analysis of Variance
Source      DF     SS      MS      F      P
Regression   1  429.910 429.910 137.92  0.000
Error       28   87.277   3.117
Total       29  517.187
```

- The R-squared value suggests that WorkExp is not an important influence factor, and that there must be other more relevant influence factors.
- WorkExp has a significant impact on Weeks.
- Although we observe a relationship between the two variables, the evidence in the data is not significant, indicated by the small p-value.
- Although we observe a relationship between the two variables, the evidence in the data is not significant, indicated by the high R-squared value.

4. A regression is performed to determine the effect of work experience (WorkExp, in years) on the number of weeks it takes unemployed people to find a new job (Weeks). 1 point

The transfer function is: Weeks = 3.81 + 0.54 WorkExp.

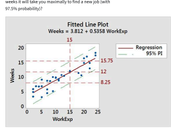
Suppose you have 10 years of work experience. What is the expected number of weeks you need to find a new job?



- Approximately 9.2.
- Cannot be determined without a prediction interval.
- Approximately 5.4.
- Approximately 11.

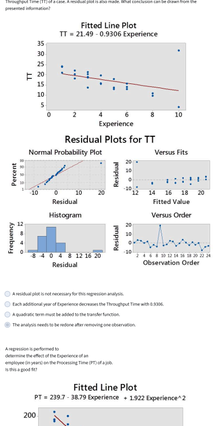
5. A regression is performed to determine the effect of work experience (WorkExp, in years) on the number of weeks it takes unemployed people to find a new job (Weeks). 1 point

Assume you have 15 years of work experience and are unemployed. What is your expectation regarding how many weeks it will take you maximally to find a new job (with 97.5% probability)?



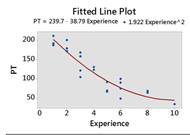
- Approximately 12 weeks.
- Approximately 15 weeks.
- Approximately 8.25 weeks.
- Approximately 15.75 weeks.

6. A regression is performed to determine the effect of the Experience of an employee (in years) on the Throughput Time (TT) of a case. A residual plot is also made. What conclusion can be drawn from the presented information? 1 point



- A residual plot is not necessary for this regression analysis.
- Each additional year of Experience decreases the Throughput Time with 0.9306.
- A quadratic term must be added to the transfer function.
- The analysis needs to be redone after removing one observation.

7. A regression is performed to determine the effect of the Experience of an employee (in years) on the Processing Time (PT) of a job. Is this a good fit? 1 point



Polynomial Regression Analysis: PT versus Experience

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The regression equation is
PT = 239.7 - 38.79 Experience + 1.922 Experience^2

S = 20.2187   R-Sq = 86.5%   R-Sq(adj) = 85.2%

Analysis of Variance
Source    DF     SS       MS      F      P
Regression  2  54856.7  27428.3  67.10  0.000
Error      21   8584.7    408.8
Total      23  63441.4

Sequential Analysis of Variance
Source    DF     SS      F      P
Linear     1  51138.0  91.44  0.000
Quadratic  1   3718.7   9.10  0.007
```

- No, a quadratic fitted line is not needed, a linear line would have been just as good.
- Yes, there is no evidence for a poor fit.
- No, there are outliers.
- No, the relationship is not significant.

8. You work at the department of a bank, that decides whether or not to grant loans for startup companies. 1 point

The CTQ is whether or not a loan is granted, and the unit is a loan request from a startup company. We consider statistically testing the following two influence factors:

1. The number of people involved with the startup project.
2. The education level (secondary school, high school, university) of the person representing the startup.

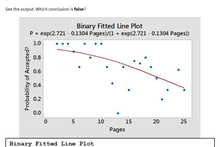
Which statistical tests are needed for this?

- 1: Logistic regression, 2: Logistic regression.
- 1: Logistic regression, 2: Chi-square analysis.
- 1: Chi-square analysis,2: Chi-square analysis.
- 1: Chi-square analysis,2: Logistic regression.

9. You work at the department of a bank, that decides whether or not to grant loans to startup companies. 1 point

You consider the influence factor "Pages", which is the number of pages of the request that is sent in. You run a statistical test to find out whether this significantly influences the decision of whether or not the loan is granted.

See the output. Which conclusion is **false**?



Binary Fitted Line Plot

```
Response Information
Variable   Value  Count
Accepted?  1         71  (Event)
           0         29
           Total    100

Deviance Table
Source      DF  Adj Dev  Adj Mean  Chi-Square  P-Value
Regression   1    15.23    15.228       15.23    0.000
  Pages      1    15.23    15.228       15.23    0.000
Error       98   105.20     1.073
Total       99   120.43
```

- The dataset contains 71 accepted requests.
- The effect of the number of pages (Pages) on the probability of acceptance is linear.
- The effect of the number of pages (Pages) on the probability of accepting a request is significant.
- The probability of accepting a request is higher for requests that only contain a few pages.

10. You work at the department of a bank, that decides whether or not to grant loans for startup companies. 1 point

You consider the influence factor Tech, which measures whether a startup operates in the technology sector, or not.

You performed a Chi-square test. Which conclusion fits the output best?

Tabulated Statistics: Tech; Accepted?

```
Rows: Tech   Columns: Accepted?

             Rejected   Accepted       All
No tech          23         33        56
               41.07      58.93    100.00
               79.31      46.48     56.00
                 23         33        56

Tech              6         38        44
               13.64      86.36    100.00
               20.69      53.52     44.00
                  6         38        44

All              29         71       100
             29.00      71.00    100.00
            100.00     100.00    100.00
                29         71       100

Cell Contents:
                   Count
                   % of Row
                   % of Column
                   % of Total
Pearson Chi-Square = 9.073; DF = 1; P-Value = 0.003
Likelihood Ratio Chi-Square = 9.542; DF = 1; P-Value = 0.002
```

- Of all rejected requests, 13.64% came from tech startups.
- Non-tech startups have a significantly higher probability of getting a grant.
- The effect of operating in the technology sector on the acceptance of grants is significant.
- Of all tech startup loan requests, 53.52% are accepted.